

Singular Value Decomposition.

Singular Vectors.

Def - First Singular Vector (right).

The first right singular vector ~~is~~ of $X \in \mathbb{R}^{n \times d}$ is defined

$$v_1 \in \mathbb{R}^d, \text{ and } v_1 = \arg \max_{\|v\|_2=1} \|Xv\|_2$$

and first singular value is $G_1 \in \mathbb{R}^+$ ~~st. $\|Xv_1\|_2 = G_1 \|v_1\|_2$~~ and $G_1 = \|Xv_1\|_2$

property.

$$v_1 = \arg \min_{\|v\|_2=1} \text{definition} \Leftrightarrow v_1 = \arg \min_{\|v\|_2=1} \sum_{j=1}^n \|x_j - P_{v_1}(x_j)\|_2^2$$

$P_{v_1}(x_j)$ ~~is the~~ $= \langle x_j, v_1 \rangle v_1$, the projected vector of x_j in the subspace spanned by v_1 .

$$X \in \mathbb{R}^{n \times d} = \begin{pmatrix} -x_1^T- \\ -x_2^T- \\ \vdots \\ -x_n^T- \end{pmatrix}, x_i \in \mathbb{R}^d$$

$$\begin{aligned} \text{Proof: } \sum_j \|x_j - P_{v_1}(x_j)\|_2^2 &= \sum_j (x_j - \langle v_1, x_j \rangle v_1)^2 = \sum_j x_j^2 + \langle v_1, x_j \rangle^2 v_1^2 - 2x_j^T v_1 \langle v_1, x_j \rangle \\ &= \sum_j x_j^2 + \sum_j \langle v_1, x_j \rangle^2 - 2 \sum_j \langle v_1, x_j \rangle^2 = \sum_j x_j^2 - \sum_j \langle v_1, x_j \rangle^2 \\ &= \sum_j x_j^2 + \sum_j \langle v_1, x_j \rangle^2 - 2 \sum_j \langle v_1, x_j \rangle^2 = C - \sum_j \langle v_1, x_j \rangle^2 = C - \|Xv_1\|_2^2 \end{aligned}$$

$$\text{thus } \arg \min_{\|v\|_2=1} \sum_j \|x_j - P_{v_1}(x_j)\|_2^2 = \arg \max_{\|v\|_2=1} \|Xv\|_2^2$$

Remarks: First singular vector is thus a line that minimizing least square error ~~that minim~~ of projected data, i.e. a best fitting line.

Def - Second Singular Vector.

$$v_2 \in \mathbb{R}^d, v_2 = \arg \min_{v_2 \perp v_1, \|v_2\|=1} \|Xv_2\|_2$$

$$G_2 \in \mathbb{R}^+, G_2 = \|Xv_2\|_2$$

Remark: v_1 and v_2 form the best fitting subspace (2 dimensional) that minimize projected ~~ls~~ least square error.

Property.

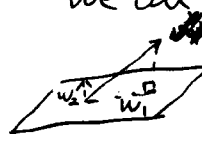
let v_1, v_2 be the first and second singular vector, let

$V = \text{span}(v_1, v_2) \subseteq \mathbb{R}^d$ be a two-dimensional subspace.

$$\begin{aligned} \text{def } V &= \arg \min_{W \subseteq \mathbb{R}^d} \sum_{j=1}^n \|x_j - P_W(x_j)\|_2^2 \\ \Rightarrow & \left\{ \begin{array}{l} W \text{ a linear subspace} \\ \dim(W) = 2. \end{array} \right\} \quad P_W: \mathbb{R}^d \rightarrow \mathbb{R}^d \text{ is projection to subspace } W. \end{aligned}$$

proof: let w_1, w_2 be two orthonormal base of W ,

we can always choose w_2 s.t. $\langle w_2, v_1 \rangle = 0$.

 (we can choose subspace that $\perp$ to v_1 , and find w_1 as intersection).

$$\sum_{j=1}^n \|x_j - (w_1 \langle w_1, x_j \rangle + w_2 \langle w_2, x_j \rangle)\|_2^2 = \sum_{j=1}^n \|x_j - P_W(x_j)\|_2^2$$

$$= \sum_{j=1}^n \langle x_j - P_W(x_j), P_W(x_j) \rangle + \sum_{j=1}^n \langle x_j - P_W(x_j), -P_W(x_j) \rangle$$

$$= \sum_{j=1}^n \|x_j\|_2^2 - 2 \sum_{j=1}^n \langle P_W(x_j), x_j \rangle + \sum_{j=1}^n \langle P_W(x_j), P_W(x_j) \rangle$$

$$\text{Since } \langle w_2, v_1 \rangle = 0 \Rightarrow P_W(x_j) = w_1 \langle w_1, x_j \rangle. \quad \langle P_W(x_j), x_j \rangle = \langle w_1, x_j \rangle^2 + \langle w_2, x_j \rangle^2$$

$$\langle P_W(x_j), x_j \rangle = \langle w_1, x_j \rangle^2 = \langle P_W(x_j), P_W(x_j) \rangle = \langle P_W(x_j), P_W(x_j) \rangle$$

$$\Rightarrow \sum_{j=1}^n \|x_j - P_W(x_j)\|_2^2 = C - \left(\sum_{j=1}^n \langle w_1, x_j \rangle^2 + \langle w_2, x_j \rangle^2 \right)$$

Since v_1 maximizes $\|x_j\|_2^2$ among all v with $\|v\|_2 = 1$, $\sum_{j=1}^n \langle w_1, x_j \rangle^2 \leq \sum_{j=1}^n \langle v_1, x_j \rangle^2$

Since v_2 maximizes $\|x_j\|_2^2$ among $v \perp v_1$ and $\|v\|_2 = 1$,

$$\sum_{j=1}^n \langle w_2, x_j \rangle^2 \leq \sum_{j=1}^n \langle v_2, x_j \rangle^2$$

thus V minimizes $\sum_{j=1}^n \|x_j - P_W(x_j)\|_2^2$ among all 2-dim subspaces.

Def r -right singular vector.

The leading r right singular vectors $v_1, \dots, v_r \in \mathbb{R}^d$ are defined as:

$$v_1 = \arg \max_{\|v\|_2=1} \|Xv\|_2, \quad v_2 = \arg \max_{\substack{\|v\|_2=1 \\ v \perp \text{span}\{v_1\}}} \|Xv\|_2 \quad \dots \quad v_r = \arg \max_{\substack{\|v\|_2=1 \\ v \perp \text{span}\{v_1, \dots, v_{r-1}\}}} \|Xv\|_2$$

singular values $\sigma_k = \|Xv_k\|_2$.

Greedy construction of best-fit subspace.

Let $v_1, \dots, v_r$ be the leading right singular vectors, $V = \text{span}\{v_1, \dots, v_r\} \subset \mathbb{R}^d$.

then $V \in \arg \min_{\substack{W \subset \mathbb{R}^d \\ \text{linear subspace} \\ \dim(W)=r}} \sum \|x_j - P_W(x_j)\|_2^2$

i.e. V is best-fit linear subspace in terms of least square error.

Singular Value Decomposition

For a rectangular matrix $X \in \mathbb{R}^{n \times d}$, $\exists$ orthogonal matrix $U \in \mathbb{R}^{n \times n}$ and $V^T \in \mathbb{R}^{d \times d}$, and a non-negative diagonal matrix $\Sigma \in \mathbb{R}^{n \times d}$ s.t.

$$X_{n \times d} = U_{n \times n} \Sigma_{n \times d} V_{d \times d}^T$$

Remarks: $\therefore$ diagonals of Σ , arrange in non-increasing order, are the singular values.

- Columns of U are the left singular vectors of X .
- Columns of V are the right singular vectors of X .

$$\begin{array}{c} \boxed{\begin{smallmatrix} d & n \end{smallmatrix}} \\ X \end{array} = \begin{array}{c} \boxed{\begin{smallmatrix} n & \dots & n \end{smallmatrix}} \\ U \end{array} \begin{array}{c} \boxed{\begin{smallmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{smallmatrix}} \\ \Sigma \end{array} \begin{array}{c} \boxed{\begin{smallmatrix} d & d \end{smallmatrix}} \\ V^T \end{array}$$

- The left singular vector $u_i = \frac{1}{\sigma_i} X v_i$, σ_i is the i th element of Σ .

Proof of Existence:

Let $X \in \mathbb{R}^{n \times d}$, and $C = X^T X \in \mathbb{R}^{d \times d}$; then C is symmetric, positive semidefinite. ~~Can~~ let the orthogonal decomposition of C as

$C = V \Lambda V^T$, where $V^T V = I$, and $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, $\lambda_i \geq 0$, n .

Consider $\bar{Z} \in \mathbb{R}^{n \times d}$ s.t. $\bar{Z}^T \bar{Z} = \Lambda$. If $\text{rank}(X) = r \leq d$, then.

π/r we can arrange Λ and V s.t. $\lambda_1, \lambda_2, \dots, \lambda_r > 0$, $\lambda_{r+1} \dots \lambda_n = 0$.

$$\text{then } \bar{Z} = \begin{pmatrix} \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_r}) & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \text{diag}(b_1, \dots, b_r) & 0 \\ 0 & 0 \end{pmatrix}$$

~~we~~ we define the left singular vectors u_i as.

$u_i = \frac{1}{b_i} X v_i$, $i=1, \dots, r$, then $\{u_i\}$ are orthonormal, as

$$u_i^T u_j = \frac{1}{b_i b_j} v_i^T X^T X v_j = \frac{1}{b_i b_j} v_i^T C v_j = \frac{1}{b_i b_j} \lambda_j \delta_{ij} = \delta_{ij}.$$

we can then extend $u_1, \dots, u_r$ s.t. ~~for any~~

$U = [u_1 | u_2 | \dots | u_n] \in \mathbb{R}^{n \times n}$ is orthogonal.

thus we have $b_i u_i = X v_i$, $i=1, \dots, r$. and $X v_i = 0$, as $X^T X v_i = C v_i = 0$

$$i > r. \Rightarrow X V = U \bar{Z} \Rightarrow X = U \bar{Z} V^T$$

Remark: This proof can be extend to $X \in \mathbb{C}^{n \times d}$, ~~then~~ with $X = U \bar{Z} V^T$

and $U \in \mathbb{C}^{n \times n}$, $V \in \mathbb{C}^{d \times d}$ unitary matrix

prop The SVD of $X = U \bar{Z} V^T$, are same as the Cereby construction.

i.e. U and $\bar{Z}$ are right singular vector and singular value.

constructed by greedy definitions.

proof: let $v'_1 := \arg\max_{\|v\|_2=1} \|Xv\|_2$ be the first right singular value

$$X = U \bar{\Sigma} V^T \text{ and } V = [v_1 v_2 \dots v_d] \in \mathbb{R}^{d \times d} \text{ and } v_i^T v_j = \delta_{ij}$$

$$\|Xv\|_2 = \|U \bar{\Sigma} U^T v\|_2 = \sqrt{v^T V \bar{\Sigma}^T U^T U \bar{\Sigma} V v} = \|\bar{\Sigma} V^T v\|_2$$

expand $v \in \mathbb{R}^d$ in basis $v_1, v_2, \dots, v_d$ as $v = \sum_{i=1}^d u_i v_i$, $\|v\|_2 = \sum (u_i)^2 = 1$

$$\|Xv\|_2 = \|\bar{\Sigma} V^T (\sum u_i v_i)\|_2 = \|\sum u_i \bar{\Sigma} V^T v_i\|_2$$

$$\left(\bar{\Sigma} V^T v_i \right)_j = \bar{\Sigma}_{jk} (V^T v_i)_k = \bar{\Sigma}_{jk} V_{kp} (v_i)_p = \bar{\Sigma}_{jk} V_{k1} v_i = \bar{\Sigma}_{j1} \delta_{ki} = \bar{\Sigma}_{j1} = \delta_{j1} G_1$$

$$= \sqrt{\|Xv\|_2^2} = \sqrt{\left\| \sum_{j=1}^d \delta_{j1} G_1 \right\|_2^2} = \sqrt{G_1^2} = G_1$$

$$\|Xv\|_2 = \sqrt{\sum_{j=1}^d (\bar{\Sigma}_{j1} u_1)^2} = \sqrt{\sum_{j=1}^d (G_j u_1)^2}, \text{ suppose } G_1 \geq G_2 \geq \dots \geq G_r$$

$$\text{noticed, } \sum_{i=1}^{\min(n,d)} G_i u_i^2 \leq G_1 \left(\sum_{i=1}^{\min(n,d)} u_i^2 \right) \leq G_1^2 \left(\sum_{i=1}^{\min(n,d)} u_i^2 \right) \leq G_1^2$$

and equality attained when $u_1 = 1 \Rightarrow v = v_1 \Rightarrow v'_1 = v_1$

Next, $v'_2 := \arg\max_{\|v\|_2=1, v \perp v_1} \|Xv\|_2$, we have the similar procedure

$$\|Xv\|_2 = \sqrt{\sum_{j=2}^d (G_j u_j)^2}, \text{ but with } u_1 = 0 \text{ (so that } v \perp v_1)$$

$$= \sqrt{\sum_{j=2}^d (G_j u_j)^2} \leq \sqrt{G_2^2} \text{ when } u_2 = 1.0 \Rightarrow v'_2 = v_2$$

Similarly, we can find all v_i to be the same as the greedy construction

Skinny SVD

$$\text{Full SVD: } X = U_{n \times n} \bar{\Sigma}_{n \times d} V_{d \times d}$$

$$\text{Notice that } \text{rank}(X) = r \leq \min(d, n) \quad \begin{pmatrix} G_1 & \dots & G_r & 0 & \dots & 0 \end{pmatrix} = \bar{\Sigma}$$

$\Rightarrow$ the singular vectors $r_1, \dots, \min(d, n)$ doesn't contribute truncated. $U_r = [u_1, \dots, u_r] \in \mathbb{R}^{n \times r}$, $V_r = [v_1, \dots, v_r] \in \mathbb{R}^{d \times r}$, $\bar{\Sigma}_r = \text{diag}(G_1, \dots, G_r) \in \mathbb{R}^{r \times r}$

$$X_{n \times d} = U_r \bar{\Sigma}_r V_r^T = \sum_{i=1}^r G_i u_i v_i^T$$

Computing SVD.

Power Method.

idea: let $C = X^T X \in \mathbb{R}^{d \times d}$, $\tilde{X} = U \tilde{\Sigma} U^T = \sum_{i=1}^r \tilde{\sigma}_i u_i u_i^T$

$$\Rightarrow C = \sum_{i=1}^r \tilde{\sigma}_i^2 u_i u_i^T \quad \text{Hence } C^2 = \sum_{i=1}^r \tilde{\sigma}_i^4 u_i u_i^T \rightarrow C^3 = \sum_{i=1}^r \tilde{\sigma}_i^6 u_i u_i^T$$

$$\rightarrow C^k = \sum_{i=1}^r \tilde{\sigma}_i^{2k} u_i u_i^T \quad \text{if } \tilde{\sigma}_1 > \tilde{\sigma}_2, \text{ then for } k \text{ large enough}$$

$$\tilde{\sigma}_1^{2k} \gg \tilde{\sigma}_2^{2k} \Rightarrow C^k \approx \tilde{\sigma}_1^{2k} u_1 u_1^T$$

$$\Rightarrow C^k \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \approx \tilde{\sigma}_1^{2k} u_1(1) u_1 \Rightarrow \frac{C^k e_1}{\|C^k e_1\|_2} = \pm u_1.$$

linear rate of convergence:

To check the convergence rate, we wish to check

$$\min \{ \|v^{(k)} - u_1\|_2^2, \|v^{(k)} + u_1\|_2^2 \} = 2 - 2|\langle v^{(k)}, u_1 \rangle|, \text{ where } v^{(k)} = \frac{C^k e_1}{\|C^k e_1\|_2} = \frac{1}{\|C^k e_1\|_2} \sum_{i=1}^r \tilde{\sigma}_i^{2k} u_i u_i^T e_1$$

$$1 = \|v^{(k)}\|_2^2 = \frac{1}{\|C^k e_1\|_2^2} \sum_{i=1}^r \tilde{\sigma}_i^{4k} | \langle u_i, e_1 \rangle |^2 \quad (u_i \text{ are orthonormal}).$$

$$= \frac{1}{\|C^k e_1\|_2^2} \left(|\langle v^{(k)}, u_1 \rangle|^2 + \sum_{i=2}^r |\langle v^{(k)}, u_i \rangle|^2 \right)$$

For the convergence, we wish $|\langle v^{(k)}, u_1 \rangle| \approx 1$, and thus the projection to other subspace is small.

$$\frac{\sum_{i=2}^r |\langle v^{(k)}, u_i \rangle|^2}{|\langle v^{(k)}, u_1 \rangle|^2} = \frac{\sum_{i=2}^r \tilde{\sigma}_i^{4k} |u_i(1)|^2}{\tilde{\sigma}_1^{4k} |u_1(1)|^2} \leq \frac{\sum_{i=2}^r \tilde{\sigma}_i^{4k}}{\tilde{\sigma}_1^{4k} |u_1(1)|^2} \leq \frac{(r-1) \tilde{\sigma}_2^{4k}}{\tilde{\sigma}_1^{4k} |u_1(1)|^2} \leq \frac{(r-1) \tilde{\sigma}_2^{4k}}{\tilde{\sigma}_1^{4k} \delta^2} \quad \text{if } |u_1(1)| \geq \delta > 0.$$

$$\Rightarrow \frac{1 - |\langle v^{(k)}, u_1 \rangle|^2}{|\langle v^{(k)}, u_1 \rangle|^2} \leq (r-1) \frac{\tilde{\sigma}_2^{4k}}{\tilde{\sigma}_1^{4k} \delta^2} \rightarrow \text{linear convergence as } \tilde{\sigma}_2 < \tilde{\sigma}_1.$$

compute C^k is costly $\rightarrow$ compute C_k instead, each step only need matrix $\times$ vector (efficient when one dim of X is small or X is sparse).

Similar linear convergence when $\tilde{\sigma}_2 > \tilde{\sigma}_1$ and $|u_1^T e_1| > \delta$.

$V^{(k)} := \frac{C^k q}{\|C^k q\|_2}$, let q expand in $\mathbb{R}^d$ orthonormal bases $\{v_1, v_2, \dots, v_n\}$
 as $q = \sum_{i=1}^d v_i G_i \Rightarrow \cancel{V^{(k)}} = \frac{1}{\|C^k q\|_2} C^k q = \frac{1}{\|C^k q\|_2} \sum_{i=1}^d G_i^{2k} v_i v_i^T \sum_{j=1}^d G_j v_j = \sum_{i=1}^d \frac{G_i^{2k}}{\|C^k q\|_2} v_i v_i^T \sum_{j=1}^d G_j v_j$

$$\text{thus } \|C^k q\|_2^2 = \sum_{i=1}^d G_i^{4k} G_i^2 \rightarrow \frac{\sum_{i=1}^d \langle V^{(k)}, v_i \rangle^2}{\|C^k q\|_2^2} = \frac{\sum_{i=1}^d G_i^{4k} G_i^2}{G_1^2 G_1^{4k}} \leq \frac{\sum_{i=1}^d G_i^{2k} G_i^2}{G_1^2 G_1^{4k}} \leq \left(\frac{G_2}{G_1}\right)^{4k} \frac{1}{\delta^2}$$

here we assume $\|q\|_2=1$ and $|C_1| = |\langle v_1, q \rangle| \geq \delta > 0$.

Note the above convergence analysis is assuming $G_2 > G_1$ strictly,
 In fact, the power methods converge to some vector in the span of singular
 vectors with "nearly highest" singular value. i.e.

let $q \in \mathbb{R}^d$ and $|\langle q, v_1 \rangle| \geq \delta > 0$, let V be the linear subspace spanned.

by v_i s.t. $G_i \geq (1-\epsilon) G_1$. let $V^{(k)} = \frac{C^k q}{\|C^k q\|_2}$ be k th power iteration,
 then it can be shown that $\|\text{Proj}_V(V^{(k)})\|_2 \leq \epsilon$, $\forall \epsilon > 0$ when $k \geq \frac{\ln(1/\epsilon\delta)}{2\epsilon}$.

proof: similar as above, we expand $q = \sum_{i=1}^r G_i v_i$ and.

$$C^k q = \sum_{i=1}^r G_i^{2k} G_i v_i. \text{ suppose } G_1, \dots, G_m \text{ satisfy } G_i \geq (1-\epsilon) G_1.$$

thus the component of $V^{(k)} \perp$ to V composed of $i = m+1, \dots, r$.

$$\|\text{Proj}_V(V^{(k)})\|_2^2 = \frac{\sum_{i=m+1}^r G_i^{2k} G_i^2}{\sum_{i=1}^r G_i^{4k} G_i^2} \leq \frac{(1-\epsilon)^{4k} G_1^{4k}}{G_1^2 G_1^{4k}} \leq \frac{(1-\epsilon)^{4k} G_1^{4k}}{\delta^2 G_1^{4k}}$$

$$\Rightarrow \|\text{Proj}_V(V^{(k)})\|_2 \leq \frac{(1-\epsilon)^{2k} G_1^{2k}}{\delta} \leq \frac{e^{-2k\epsilon}}{\delta} \leq \epsilon \text{ when } k \geq \frac{\ln(1/\epsilon\delta)}{2\epsilon}.$$

Remark: q can be chosen as random from $q \sim \mathcal{N}(0, I_d)$.

• when chosen random by, $|\langle q, v_1 \rangle| \geq \delta > 0$ can be shown satisfied
 with high probability.

Subspace Power Method.

finding right singular values are same as finding eigen vector for $X^T X$, can use orthogonal iteration algorithm for that.
Let $M \in \mathbb{R}^{d \times d}$ be a positive semidefinite with eigen decomposition
$$M = V \Lambda V^T = \sum_{i=1}^d \lambda_i v_i v_i^T$$

Initialize $Q_0 \in \mathbb{R}^{d \times L}$ a random L -dimensional subspace of $\mathbb{R}^d$.

Repeat till convergence

$$t+1 \leftarrow t.$$

$$Z_{t+1} = M \cdot Q_t$$

$Q_{t+1} = QR[Z_{t+1}]$, where $QR(Z) := Q$ when $Z = QR$ as QR decomposition.

Q then represent the first L right singular vectors

Best rank- k Approximation

Recap on matrix norm

Frobenius norm $\|\cdot\|_F$: $\|X\|_F = \sqrt{\sum_{i,j} x_{ij}^2} = \sqrt{\sum_{i=1}^r b_i^2}$ singular values.

Schatten p -norm $\|\cdot\|_p$: operate on the singular values.

$X \in \mathbb{R}^{n \times d}$, $r \leq \min(n, d)$ be $\text{rank}(X)$

$$\|X\|_p := \left(\sum_{i=1}^r b_i^p(X) \right)^{1/p}, \quad 1 \leq p < \infty$$

• The spectral norm $\|\cdot\|_2 = \|\cdot\|_{s_0} = b_1$ and $\|\cdot\|_F = \|\cdot\|_{s_2}$.

$$\|X - X_k\|_p = \min_{B: \text{rank}(B) \leq k} \|X - B\|_p = \left(\sum_{i=k+1}^r b_i^p \right)^{1/p}$$

let $X \in \mathbb{R}^{n \times d}$, $\text{rank}(X) = r \leq \min\{n, d\}$, we define k -truncated SVD of X as k -truncated SVD.

$$X_k = \sum_{i=1}^k \sigma_i u_i v_i^T, \quad k \in \{1, 2, \dots, r\}. \quad \text{where } X = \sum_{i=1}^r \sigma_i u_i v_i^T \text{ is the full SVD.}$$

Then X_k is the "best" rank- k approximation in terms of orthogonally invariant norm. $\|\cdot\|$ i.e. as

$$X_k = \arg \min_{B \in \mathbb{R}^{n \times d}, \text{rank}(B) \leq k} \|A - B\|$$

i.e. X_k is the most 'close' rank- k (w.r.t) matrix to X in terms of orthogonally invariant norm.

~~$\| \cdot \|$ is~~ $\|O_1 X O_2\| = \|X\|$, $\forall X \in \mathbb{R}^{n \times d}$, $O_1 \in \mathbb{R}^{n \times n}$, $O_2 \in \mathbb{R}^{d \times d}$ are both orthogonal.

Example: • Frobenius norm $\|\cdot\|_F$
• Spectral norm $\|\cdot\|_2$.

Proof for spectral norm:

let $X \in \mathbb{R}^{n \times d}$, $\text{rank}(X) = r \leq \min\{n, d\}$, $X = \sum_{i=1}^r \sigma_i u_i v_i^T$ and $X_k = \sum_{i=1}^k \sigma_i u_i v_i^T$
 $\|X - X_k\|_2 = \left\| \sum_{i=k+1}^r \sigma_i u_i v_i^T \right\|_2 = \sigma_{k+1}$, we hope to show $\|A - B\|_2 \geq \sigma_{k+1}$

Consider $w \in \mathbb{R}^d$ with unit norm $\|w\|_2 = 1$. we can choose $w \in \text{Null}(B)$, i.e. $Bw = 0$, on the other hand, we can choose w also in $\text{span}\{v_1, \dots, v_m\}$, in this case $\|Aw\|_2 = \sqrt{\sigma_1^2 + \dots + \sigma_k^2} \geq \sigma_{k+1}$

thus if $w \in \text{Null}(B) \cap \text{span}\{v_1, \dots, v_m\}$, then.

$$\|A - B\|_2 = \max_{v \in \mathbb{R}^d, \|v\|_2=1} \|(A-B)v\|_2 \geq \|(A-B)w\|_2 = \|Aw\|_2 \geq \sigma_{k+1}.$$

Proof of Frobenius norm.

Consider $\|X - B\|_F$, where $\text{rank}(B) = p \leq k \leq \text{rank}(X) = r$.

denote $B = \begin{pmatrix} -b_1^T \\ -b_2^T \\ \vdots \\ -b_n^T \end{pmatrix}$ and $X = \begin{pmatrix} -x_1^T \\ -x_2^T \\ \vdots \\ -x_n^T \end{pmatrix} \Rightarrow \|X - B\|_F^2 = \|x_1 - b_1\|_2^2 + \dots + \|x_n - b_n\|_2^2$

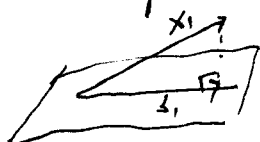
let $\{v_1, \dots, v_p\}$ be the orthogonal, and form a base in the subspace of B , and extend to $\{v_1, \dots, v_p, \dots, v_r\}$ an orthonormal bases in X .

then we can expand

$$b_i = \beta_1' v_1 + \beta_2' v_2 + \dots + \beta_p' v_p, \quad x_i = x_1' v_1 + x_2' v_2 + \dots + x_r' v_r.$$

$$\Rightarrow \|b_i - x_i\|_2^2 = (\beta_1' - x_1')^2 + \dots + (\beta_p' - x_p')^2 + \dots + x_r'^2.$$

which minimized when b_i is the projection of x_i to the subspace. i.e. $\beta_1' = x_1', \beta_2' = x_2', \dots, \beta_p' = x_p'$. This can be visualized



in 3D case where the distance between two vectors is minimized when b_i is the projection of x_i to the plane.

Since we are looking for all subspace V of dimension at most k , there could be a specific subspace that is better.

In previous, we have claim that the greedy construction of SVD singular vector minimize the distance between the vector to orthogonal subspace. Thus $\{v_1, \dots, v_k\}$ should be the first k singular vectors. the projection matrix of X to this is

$$X_1^k = (x_1^T v_1) v_1 + (x_2^T v_1) v_2 + \dots + (x_n^T v_1) v_1$$

$$X_n^k = (x_n^T v_1) v_1 + \dots + (x_n^T v_k) v_k$$

$$\Rightarrow \begin{pmatrix} -x_1^T v_1 \\ -x_2^T v_1 \\ \vdots \\ -x_n^T v_1 \end{pmatrix} = \sum_{i=1}^k v_i v_i^T X = \sum_{i=1}^k G_i U_i V_i^T = X_k$$

$$\Rightarrow \|X - B\|_F \geq \|X - P_W(X)\|_F \geq \|X - P_W^k(X)\|_F \geq \|X - X_k\|_F$$